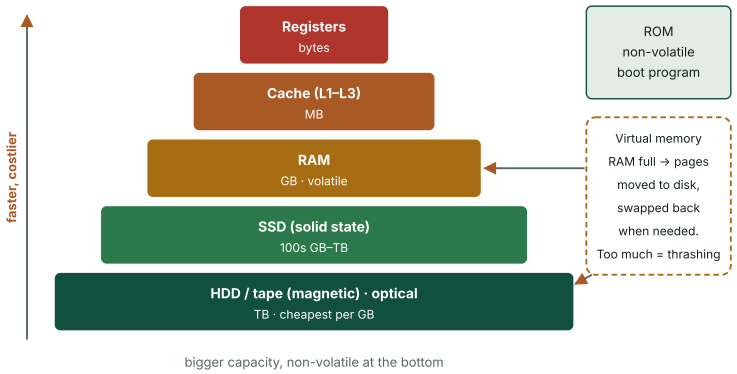
Main memory, cache and secondary storage

FD10 FACT FILE

Specification: students should be able to

- Describe main memory, distinguishing between RAM and ROM, and explain the uses of each
- Describe the purpose of cache memory, how a cache controller uses it, and the levels of cache
- Explain how virtual memory is used and its effect on performance
- Explain the need for secondary storage and the factors that influence the choice of storage device
- Describe magnetic, optical and flash (solid-state) secondary storage media, including how data is stored and accessed
- Evaluate the use of secondary storage media for common applications

Key words: main memory, random access memory (RAM), read-only memory (ROM), volatile, non-volatile, cache memory, cache controller, virtual memory, disk thrashing, secondary storage, magnetic storage, tracks and sectors, serial access, direct access, optical storage, flash memory, solid-state drive (SSD)



The memory hierarchy from registers to magnetic and optical storage: faster and more expensive at the top, larger at the bottom, with virtual memory linking RAM and secondary storage

Memory and storage in overview

A computer uses several kinds of storage, arranged as a hierarchy. At the top are the registers and cache, which are tiny, extremely fast and costly per byte. Next is main memory, the store the processor can address directly, which holds the programs and data in use. At the bottom is secondary storage, which is much slower but far larger and cheaper, and which keeps its contents when the power is off. The design aims to give the processor the speed of the fast levels while providing the capacity of the slow ones, by moving data up the hierarchy just before it is needed. When you open a file, it is copied from secondary storage into RAM, and the parts the processor is using are copied again into cache. Main memory consists of RAM, which is volatile and can be read and written, and ROM, which is non-volatile and holds fixed programs. Secondary storage comes in three main technologies: magnetic, such as hard disks and tape; optical, such as CDs, DVDs and Blu-ray discs; and flash, such as SSDs, USB sticks and memory cards. Choosing the right device means weighing speed, capacity, durability, portability, reliability and cost against what the user needs.

RAM and ROM

Random access memory holds the operating system, applications and data that are currently in use. The processor can read from it and write to it, and any location can be reached directly in the same time. It is volatile, so everything in it is lost when the power is switched off, which is why programs must be loaded into RAM each time they are used and why work must be saved to secondary storage. RAM is measured in gigabytes, and because secondary storage is so much slower, having more RAM generally makes a computer more responsive: more programs and data can stay in memory rather than being fetched repeatedly from disk. Read-only memory is non-volatile and its contents are fixed when it is made, so the user cannot change them in normal use. On a PC it holds the firmware that runs when the computer is switched on, testing the hardware and loading the operating system from secondary storage, which has to be in ROM because RAM is empty at that point. ROM is also used in embedded systems, such as the controller in a dishwasher or central heating timer, where the program never needs to change. Many modern devices use rewritable flash memory for this firmware so that it can be updated.

Cache memory

Even though RAM is much faster than a disk, the processor can still work faster than RAM can supply it, so it would spend time waiting. Cache memory solves this by keeping copies of data and instructions in a small, very fast memory on or right next to the processor. Cache is made from a faster and more expensive type of memory than RAM, so a processor with more cache costs more. It has two jobs: it keeps instructions and data that are used often, and it reads ahead. A cache controller manages the transfer of data from RAM into the cache. When the processor needs an item, it first checks the cache. If the item is there, a cache hit, it is supplied immediately. If not, a cache miss, the item is fetched from RAM, placed in the cache and then used, and the controller may also fetch the neighbouring addresses, since program instructions usually follow one another in memory. Cache is organised in levels: level 1 is the smallest and fastest and is built into each core, level 2 is larger and slightly slower, and level 3 is larger again and usually shared by all the cores on the chip.

Virtual memory

When so many programs are open that RAM is full, the operating system can use an area of secondary storage as if it were extra RAM. This is virtual memory. Memory is managed in fixed-size blocks called pages. Pages that are not being used at the moment are copied out of RAM to the disk, freeing space for the pages that are needed now. When a page held on the disk is needed again, it is copied back into RAM, which may mean another page has to be moved out to make room. Virtual memory lets a user run more programs, or larger ones, than the physical RAM would otherwise allow, and it stops the computer simply refusing to open another program. The cost is speed. Even an SSD is far slower than RAM, and a hard disk is slower still, so every swap holds the processor up. If RAM is badly overloaded, the computer can spend more time moving pages back and forth than doing useful work, a condition called disk thrashing, in which the system becomes very sluggish and the disk activity light may stay on. The usual cure is to fit more RAM or close some programs, which reduces how often virtual memory is needed.

Magnetic and optical storage

A hard disk drive contains one or more rigid platters coated with magnetic material, mounted on a spindle inside a sealed case. Each surface is organised into concentric tracks, which are divided into sectors. The platters spin at thousands of revolutions per minute while read/write heads on an actuator arm move across them. To write, the operating system looks up its file system table for free space, the data is passed to the disk controller, the arm moves the head to the right track, and when the required sector spins beneath it, tiny regions are magnetised in one of two directions to represent 0s and 1s. Hard disks give direct access, huge capacity and a low cost per gigabyte, but their moving parts can be damaged by a knock. Magnetic tape stores data serially, so the drive must wind through the tape to reach a file. It is slow to search but very cheap for its vast capacity, so it suits archives and backups. Floppy disks used similar principles but are obsolete. Optical discs store data as marks on a reflective layer that a laser reads. CDs, DVDs and Blu-ray discs come as ROM, recordable (R) and rewritable (RW) types; they are cheap and portable but slow and easily scratched.

Flash memory and SSDs

Flash memory stores data as electrical charges held in semiconductor cells. It is non-volatile and has no moving parts, which makes it fast, quiet and resistant to being dropped or shaken. USB memory sticks and memory cards, such as the SD and microSD cards used in cameras, drones and phones, are small, light and portable. When a USB stick is plugged in, the operating system loads a suitable driver, reads the stick's file table to learn its folder structure, and then, when a file is opened, sends the required address to the device, which returns the data. A solid-state drive uses the same technology but is designed to sit inside the computer in place of a hard disk. Compared with a hard disk, an SSD gives much faster access because there is no waiting for a head to move or a platter to spin; it uses less power, which extends laptop battery life; it is smaller and lighter, suiting thin laptops and tablets; and it survives knocks better. Its disadvantages are a higher cost per gigabyte and the fact that each cell can be rewritten only a limited number of times, although modern drives spread writes evenly to prolong their life. Falling prices mean SSDs are now the normal main drive in most new laptops.

Choosing storage for an application

Every storage question comes down to matching the medium to the need. Speed matters when data must be opened instantly, as with a booking system or the drive holding the operating system, which favours an SSD. Capacity matters for large collections of video, images or records, which favour hard disks or, for archives, tape. Durability and reliability matter in harsh conditions: a camera on a racing car, a field-research tablet or a delivery driver's handheld needs flash storage with no moving parts. Portability matters when data has to travel, which favours USB sticks, memory cards and optical discs. Cost per gigabyte matters for bulk storage and backups, where tape and hard disks are cheapest. The type of access needed is also important: tape is serial, so it is a poor choice when individual files are needed often, but fine for a backup that will only be restored in an emergency. Whether data must be changed matters too: a write-once disc protects a distributed file from alteration, while a rewritable medium suits work in progress. Increasingly, organisations also use cloud storage and network drives, which add off-site protection and sharing but depend on an internet connection and on trust in the provider.

T Key terms from the mark schemes

Main memory
The memory that the processor can address directly, holding the programs and data currently in use (RAM and ROM).
`Credited: Primary memory / immediate access store`

RAM
Random access memory: volatile read/write main memory that holds the operating system, programs and data currently in use.

ROM
Read-only memory: non-volatile main memory whose contents are fixed when made, used for start-up firmware and embedded programs.

Volatile
Describes memory that loses its contents when the power is switched off.

Cache memory
Small, very fast and expensive memory on or beside the processor that holds frequently used data and instructions and data read ahead from RAM.
`Credited: CPU memory`

Cache controller
The device that transfers data from RAM into cache, predicting and pre-fetching the data the processor is likely to need next.

Virtual memory
An area of secondary storage used by the operating system as an extension of RAM when RAM is full, with pages swapped in and out.

Disk thrashing
Severe loss of performance when the computer spends more time swapping pages between RAM and virtual memory than processing.

Secondary storage
Non-volatile storage that keeps programs and data permanently when the power is off, e.g. hard disk, SSD, optical disc.
`Credited: Backing storage / auxiliary storage`

Serial access
Data must be read in order from the start until the required item is reached, as on magnetic tape.
`Credited: Sequential access`

Optical storage
Storage in which a laser writes and reads marks on the reflective surface of a disc, e.g. CD, DVD and Blu-ray.

Solid-state storage
Non-volatile storage using flash memory with no moving parts, e.g. SSDs, USB drives and memory cards.
`Credited: Flash memory`

Exam focus

Definition questions on RAM and ROM usually award one mark per characteristic, so state volatility, whether it can be written to, and what it is used for, and make the difference explicit. Avoid saying ROM stores the operating system on a PC: it stores the start-up firmware that loads the operating system. Cache answers should mention speed, location near the processor, frequently used data and reading ahead, and a diagram question expects the CPU, cache, cache controller and RAM with the hit and miss paths. Describe virtual memory as a process: RAM full, pages moved to secondary storage, swapped back when needed, slower performance. Storage scenario questions often say 'do not use the same device or reason twice', and marks come mainly from the justification, which must be linked to the scenario rather than generic. Use the factors examiners credit: speed, capacity, durability, portability, reliability and cost. 'Fast' alone earns little; 'an SSD has faster access than a hard disk because it has no moving parts' earns more. Remember that a CD-ROM drive cannot write data, that tape gives serial access, and that SSD advantages include low power, speed, robustness and small size.

? Check yourself

1. State two differences between RAM and ROM. [2]
2. Explain why the start-up instructions of a PC are stored in ROM rather than RAM. [2]
3. Using a diagram, explain how the CPU makes use of cache memory when it needs an instruction. [4]

Answers: 1. Any two from: RAM is volatile, ROM is non-volatile (1) / RAM can be read and written, ROM can only be read (in normal use) (1) / RAM holds programs and data currently in use, ROM holds fixed start-up programs/firmware (1) / RAM is usually much larger than ROM (1) 2. RAM is volatile so it is empty when the computer is switched on (1) / ROM is non-volatile so the start-up instructions are available immediately to load the operating system (1) 3. Diagram showing CPU, cache, cache controller and RAM (1) / CPU first checks the cache for the instruction (1) / if present (cache hit), it is used directly, having been pre-fetched by the cache controller (1) / if not (miss), it is fetched from RAM, placed in cache and then processed; controller may read further data ahead (1)